In other words, the wave function, $\psi(x)$, of a particle moving in the x direction is the projection of a state-vector $|\Psi\rangle$ onto the eigenvectors of position. We will also refer to $\psi(x)$as the *wave function in the position representation*.

8.2.2 Momentum and Its Eigenvectors

Position is intuitive; momentum is less so, particularly in quantum mechanics. It will only be later that we see the connection between the operator that we identify with momentum and the familiar classical concept of mass times velocity. But I assure you that we will make the connection.

For now, let's take the abstract mathematical route. The momentum operator in quantum mechanics is called **P**, and it is defined in terms of the operator $-i\mathbf{D}$:

$$-i\mathbf{D} = -i\frac{d}{dx}.$$

As we saw earlier in Eq. 8.10, we need the factor $-i$ to make this operator Hermitian.

We could just define **P** to be $-i\mathbf{D}$, but if we did, we would run into a problem later when we connect these ideas to those of classical physics. The reason should be clear—there's a dimensional mismatch. In classical physics, the units of momentum are mass times velocity—in other words, mass times length divided by time (ML/T). On the other hand, the operator **D** has units of inverse length, or $1/L$.The resolution of the mismatch is provided by Planck's constant $\hbar$, which has units of ML^2/T. The correct relation between **P** and **D** is therefore

$$\mathbf{P} = -i\hbar\mathbf{D} \tag{8.14}$$

or, in terms of its action on wave functions,